

Regression for STA2020F

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Abstract

These notes provide a summary of the content covered in the regression section of the STA2020F 2019 course. The purpose of these notes is to provide an overview of the regression core material in a condensed format to assist in the revision of the concepts - reinforcing what is taught in more detail in the course lectures and what is described in the regression section slides. In addition, relevant `RStudio` code and examples are shown.

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1. Correlation

Introduction

Regression and correlation analyses are techniques for quantifying the relationship between two or more continuous variables. You make this kind of relationships in your head all the time, for example when you calculate the age of a child based on her height, you are assuming the older she is, the taller she will be. We shall concentrate on situations in which the general trend is linear; that is, as one variable changes the other variable follows on the average a trend which can be represented by a straight line. Our objectives in the Correlation section are to learn about:

1. Visualising data using a scatter-plot.
2. The definition and properties of the correlation coefficient.
3. Estimating and interpreting the correlation coefficient.

Scatter Diagrams

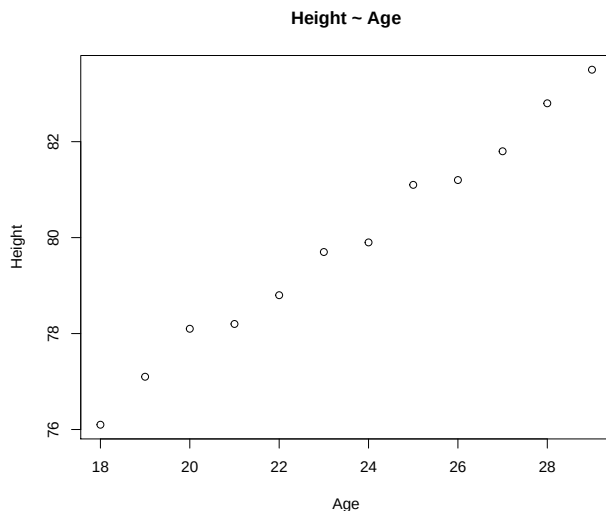
A first step in any correlation/regression analysis should be a plot of the data. The basic graphical technique for the two-variable situation is the scatter diagram. In general the data refer to a number of individuals, each of which provides observations on two variables. In the scatter diagram each variable is allotted one of two co-ordinate axes and each observation defines a point, of which the co-ordinates are the observed values of the two variables. The scatter diagram gives a compact illustration of the relationship between the two variables. For example we may want to quantify the relationship between the age of a child (months) and a child's height (cm):

Table 1. Age and Height data

Age	Height
18	76.1
19	77.1
20	78.1
21	78.2
22	78.8
23	79.7
24	79.9
25	81.1
26	81.2
27	81.8
28	82.8
29	83.5

In RStudio we input the data and generate the plots using:

```
Age = c(18, 19, 20, 21, 22, 23, 24, 25,
        26, 27, 28, 29)
Height = c(76.1, 77.1, 78.1, 78.2, 78.8,
           79.7, 79.9, 81.1, 81.2, 81.8, 82.8, 83.5)
plot(Age, Height, main="Height ~ Age")
```



We observe that there is positive linear relationship and proceed to quantify it.

Correlation coefficient

The strength of the linear association between two variables y and x can be summarised by the **sample correlation coefficient**, r_{xy} . For n pairs of observations $(y_i, x_i), i = 1 \dots n$, the correlation coefficient is defined by:

$$r_{xy} = \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i^n (x_i - \bar{x})^2 \sum_i^n (y_i - \bar{y})^2}} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

where $\bar{x}$ = the mean of the n values for x and $\bar{y}$ = the mean of the n values for y . Note that the correlation coefficient is a dimensionless quantity between -1 and +1. In RStudio this correlation coefficient is estimated using

```
cor(Age, Height)
## [1] 0.9942309
```

In this course, we do not perform any hypothesis tests related to r_{xy} .

2. Simple Linear Regression

Introduction

Simple Linear Regression is a technique for describing the relationship between the values of two continuous variables that enables us to predict the value of one variable for an individual when we only know the value of the other variable. **Linear** regression refers to a **straight line** relationship between variables. **Simple** linear regression refers to the situation where there is only **one** explanatory/independent variable. The method fits a straight line to the data that in some sense gives the “best” prediction of y for any value of x . This will be a line that minimises the distance between the data and the fitted line. Our objectives in the Simple Linear Regression section are to learn:

1. The parameters of the mathematical model for such a relationship
2. How to estimate the parameters of this model using RStudio
3. How to interpret the meanings of the estimated parameters
4. How to test if the relationship is statistically significant
5. How to predict values of y given a new value of the explanatory variable x .
6. How to set up confidence intervals for the estimates and predicted values.

The Linear Model

We assume

$$y = \beta_0 + \beta_1 x + \varepsilon \text{ where } \varepsilon \sim N(0, \sigma^2)$$

In this expression:

- $y \implies$ Response /Dependent variable,
- $\beta_0 \implies$ Intercept/Expected value of y when $x = 0$,
- $\beta_1 \implies$ Regression coefficient/Slope of the line/ Amount of change in y for a 1 unit change in x ,
- $x \implies$ Predictor/Independent variable,
- $\varepsilon \implies$ Error term (since not all points will lie on the line exactly).

Estimates are derived using RStudio as shown in Figure 1.

```

> mod <- lm(Height ~ Age)
> summary(mod)

Call:
lm(formula = Height ~ Age)

Residuals:
    Min       1Q   Median       3Q      Max
-0.28333 -0.24583  0.00303  0.15985  0.45303

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  65.01061    0.51197   126.98 < 2e-16 ***
Age           0.63182    0.02155    29.31 4.98e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2578 on 10 degrees of freedom
Multiple R-squared:  0.9885,    Adjusted R-squared:  0.9873
F-statistic: 859.2 on 1 and 10 DF,  p-value: 4.984e-11

```

Figure 1. Estimating parameters in RStudio

Estimating parameters

The model estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ which are sample estimates of the linear regression population parameters. Estimates are calculated to produce the line of best fit $\hat{y}$. The derived line is the one which minimizes the sum of the squared residuals which are each calculated using

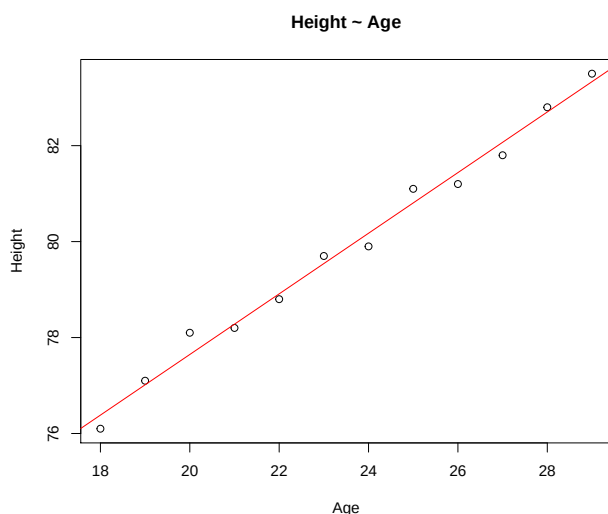
$$e_i = \hat{y}_i - y_i$$

and quantify the difference between the observed and predicted values.

```

plot(Age, Height, main="Height ~ Age")
abline(mod, col = "red")

```



For the Height and Age example:

- $\hat{\beta}_0 = 65.01 \implies$ The expected (average) height of a child of age 0 is 65.01cm
- $\hat{\beta}_1 = 0.63 \implies$ The average increase in the height of a child for every monthly increase in age 0.63cm

Hypothesis Testing

The goal of hypothesis testing is to draw conclusions about a population parameter using the information from a sample of data. Basic steps of Hypothesis Testing are:

- Define the null hypothesis, H_0
- Define the alternative hypothesis, H_a
- Define a significance level, α , usually 0.05
- Calculate the test statistic
- Calculate the p-value (probability of getting a result 'as or more extreme' than that observed if H_0 is true)
- If the p-value is $\leq \alpha$, reject H_0 . Otherwise, fail to reject H_0

For linear regression, the test statistics required for hypothesis testing are linked to the Analysis of Variance (ANOVA) table shown in Figure 2.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression	1	$SSR = \sum_i (\hat{y}_i - \bar{y})^2$	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE}$
Error	$n - 2$	$SSE = \sum_i (\hat{y}_i - y_i)^2$	$MSE = \frac{SSE}{n - 2}$	
Total	$n - 1$	$SST = \sum_i (y_i - \bar{y})^2$	$MST = \frac{SST}{n - 1}$	

Figure 2. Simple Linear Regression ANOVA

ANOVA table

The total variation in y with no knowledge of x is given by SST ¹ and we can break this into two parts:

$$\underbrace{\sum (y_i - \bar{y})^2}_{\substack{\text{Total SS} \\ n-1 \text{ d.o.f}}} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{\substack{\text{Error SS} \\ n-2 \text{ d.o.f}}} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{\substack{\text{Regression SS} \\ 1 \text{ d.o.f}}}$$

The larger the proportion of SST that is made up of SSR, the better the regression line is doing in terms of modelling the variability in y . The mean (average) square for each of the sum of squares expressions is given by:

$$MS = \frac{SS}{d.o.f}$$

For example,

$$MSE = \frac{SSE}{n - 2} = s_e^2$$

which estimates σ^2 , the variability of y about the line, or equivalently, the variance of the error terms.

Test for overall significance of linear regression

We can test for the significance of the regression,

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

using the test statistic:

$$F_{stat} = \frac{MSR}{MSE} \sim F_{1, n-2}$$

¹By definition, this is the same as SS_y

Another way of quantifying the degree of success of the model is to determine the proportion of the total variation in the dependent variable that is accounted for by the regression. This quantity, known as the Multiple R^2 or the Coefficient of Determination, is denoted by R^2 and equals SSR/SST . It can be shown that for simple linear regression this is in fact the square of the correlation coefficient, r_{xy} .

Tests for individual regression coefficients

The residual standard error $s_e = \sqrt{MSE}$ can be used to determine standard errors for the coefficients which can in turn be used to derive test statistics. Note:

$$s_{\beta_1} = \frac{s_e}{\sqrt{SS_x}} \text{ and } s_{\beta_0} = s_{\beta_1} \sqrt{\sum \frac{x_i^2}{n}}$$

We can test for the significance of β_j for $j \in [0, 1]$,

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0$$

using the test statistic:

$$t_{val} = \frac{\hat{\beta}_j}{s_{\beta_j}} \sim t_{n-2}$$

Confidence intervals for β_j

The $\beta_j 100(1 - \alpha)\%$ confidence intervals are:

$$\hat{\beta}_j \pm t_{\frac{\alpha}{2}, n-2} s_{\beta_j}$$

For example, we can now calculate the 95% confidence interval for the slope in our height and age analysis. Given that $t_{0.025, 10} = 2.23$,

$$\beta_1 \in 0.63182 \pm 2.23 * 0.02155 = [0.58; 0.68]$$

which excludes 0 $\implies$ **statistically significant!**

Confidence intervals for the prediction

A $100(1 - \alpha)\%$ confidence interval for the **mean** y for a given value of x_p is given by

$$y \in \hat{y} \pm t_{a/2, n-2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_x}}$$

A $100(1 - \alpha)\%$ prediction interval for the **point estimate of or individual value of** y for a given value of x_p is given by

$$y \in \hat{y} \pm t_{a/2, n-2} s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_x}}$$

In general, the values of SS_x and $\bar{x}$ will be given to you. For our Height and Age analysis we can use `RStudio` to calculate them. Let us predict the height of a child who is 30 months old and determine the 95% prediction interval for the point estimate. Recall, $t_{0.025, 10} = 2.23$.

```
(Prediction = 65.01061 + 0.63182*30)

## [1] 83.96521

(xb=mean(Age)); (SSx=sum((Age-xb)^2))

## [1] 23.5
## [1] 143

sc = sqrt(1+1/12+((30-xb)^2)/SSx)

(Lower= Prediction-2.23*0.2578*sc)

## [1] 83.29016

(Upper= Prediction+2.23*0.2578*sc)

## [1] 84.64026
```

Underlying assumptions

There are three underlying assumptions for the linear regression model:

1. The relationship between x and y is linear,
2. The model errors are independent
3. The errors $\varepsilon \sim N(0, \sigma^2)$ i.e. they are normally distributed with a mean of zero and a constant variance

3. Multiple (Linear) Regression**Introduction**

Multiple Regression is a technique for describing the relationship between a continuous response y and a set of more than one explanatory variables $x_j, j = 1 \dots p$, say. The simple linear regression model can be extended as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

We still assume $\varepsilon \sim N(0, \sigma^2)$. Our objectives in the Multiple Regression section are to learn:

1. How to estimate the parameters of this model using `RStudio`
2. How to interpret the meanings of the estimated parameters
3. How to predict values of y given a new values of the explanatory variables
4. How to test if the relationships are statistically significant
5. How to set up confidence intervals for the estimates
6. The concept of multicollinearity and its potential effect on inference
7. How to include and interpret categorical variables and interactions

Example dataset

As an example, a researcher has collected data on 600 high school students. She is interested in predicting a psychological variable, **motivation** score, using academic tests scores for **read**, **write**, and **science**, as well as a *categorical* variable **program** giving the type of program the student is in (general, academic, or vocational).

The estimated model

As with simple linear regression, the model estimates the regression parameters for the sample to fit the equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p$$

For the motivation example, the data are read into `RStudio` and a multiple regression is performed for the academic scores as shown in Figure 3:

```

> mr <- read.csv("MR_eg.csv")
> attach(mr)
> mrmod <- lm(motivation ~ read + write + science)
> summary(mrmod)

Call:
lm(formula = motivation ~ read + write + science)

Residuals:
    Min       1Q   Median       3Q      Max
-42.197 -10.148  -0.150   9.871  48.839

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.53083    3.77165   8.095 3.24e-15
read         0.22544    0.09058   2.489  0.0131
write        0.36051    0.08275   4.357 1.55e-05
science     -0.18382    0.08918  -2.061  0.0397
---

```

Figure 3. Multiple regression in RStudio

The interpretation of the variables follows the same logic as with simple linear regression but with the acknowledgement of the presence of other variables in the model. For the motivation example:

- $\hat{\beta}_0 = 30.53 \implies$ The expected motivation score for a student who obtained 0 for all their academic scores is 0.
- $\hat{\beta}_{read} = 0.23 \implies$ The average increase in the motivation score for every unit increase in the read score is 0.23, holding the write and science scores constant.
- $\hat{\beta}_{write} = 0.36 \implies$ The average increase in the motivation score for every unit increase in the write score is 0.36, holding the read and science scores constant.
- $\hat{\beta}_{science} = 0.18 \implies$ The average decrease in the motivation score for every unit increase in the science score is 0.18, holding the read and write scores constant.

Dummy variables

So far we have assumed quantitative continuous explanatory variables e.g. temperature, distance and age. In many regression analyses, the variables are qualitative in nature e.g. sex (male or female), colour (black, white), nationality or employment status (employed, unemployed). These are defined on a nominal scale and are referred to as **categorical** variables. They have no intrinsic ordering to their constituent categories or natural scale of measurement.

A **dummy** variable is an artificial variable created to represent a categorical variable. It is a binary (0 or 1) variable constructed by assigning values based on the presence of a category attribute, where “1” indicates the presence of the attribute and “0”, its absence. For example, “1” can indicate male and “0” female.

For a categorical variable with k categories, $k - 1$ dummy variables are created e.g. if a categorical variable has 3 categories - “A”, “B” and “C” then one of the categories is arbitrarily chosen to be the reference, say “A” and then two dummy variables are created. The first will take on a value of 1 if the category is “B” and 0 otherwise and the other will take on a value of 1 if the category is “C” and 0 otherwise, with “A” being 0 in both cases.

In our motivation dataset, the *program_* variable is categorical variable with three categories. Regression of *motivation* against *write* and *program_* yields the output shown in Figure 4:

```

> mod2 <- lm(motivation ~ write + program_)
> summary(mod2)

Call:
lm(formula = motivation ~ write + program_)

Residuals:
    Min       1Q   Median       3Q      Max
-44.602  -9.285  -0.621   9.717  42.796

Coefficients:
            Estimate Std. Error t value
(Intercept)  33.31112    3.27395  10.175
write        0.35086    0.06063   5.787
program_general -7.03377    1.49989  -4.690
program_vocational  4.80116    1.35833   3.535
---

```

Figure 4. Dummy variable example

The inclusion of the dummy variables changes the interpretation of the intercept. For our example we note

- $\hat{\beta}_0 = 33.31 \implies$ The expected motivation score for a student who obtained 0 for their write score and who is in the academic program (the reference) is 33.12.
- $\hat{\beta}_{write} = 0.35 \implies$ The average increase in the motivation score per unit increase in the

write score is 0.35, holding the program constant.

- $\hat{\beta}_{\text{program_general}} = -7.03 \implies$ The average decrease in the motivation score comparing general to academic program students is 7.03, holding the write score constant.
- $\hat{\beta}_{\text{program_vocational}} = 4.80 \implies$ The average increase in the motivation score comparing vocational to academic program students is 4.80, holding the write score constant.

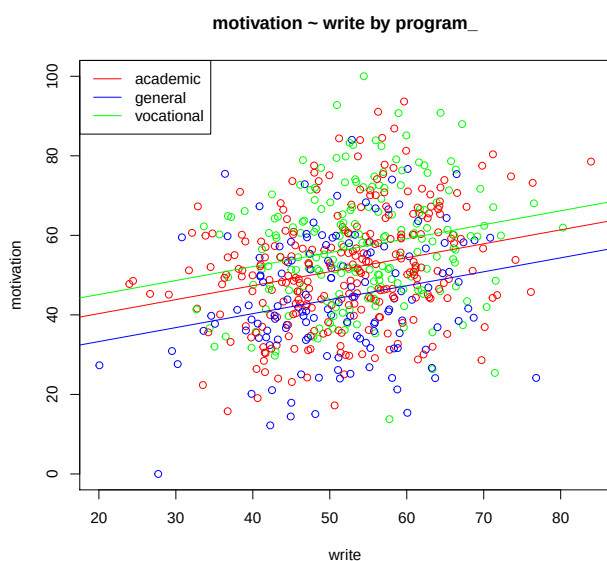
This means that if we want to predict the motivation score (y) using the write score (x) and academic program, there are 3 possible estimation equations for the academic (1), general (2) and vocational (3) programs respectively:

$$(1)\hat{y} = 33.31 + 0.35 * x$$

$$(2)\hat{y} = 33.31 + 0.35 * x - 7.03$$

$$(3)\hat{y} = 33.31 + 0.35 * x + 4.80$$

```
coef <- coefficients(mod2)
plot(write, motivation,
     main="motivation ~ write by program_",
     col=c("red", "blue", "green")[program_])
abline(coef[1], coef[2], col="red")
abline(coef[1]+coef[3], coef[2], col="blue")
abline(coef[1]+coef[4], coef[2], col="green")
legend("topleft",
     legend=c("academic", "general", "vocational"),
     col=c("red", "blue", "green"), lty=1)
```



These produce 3 parallel lines suggesting that across the programs, expected motivation scores differ but

changes in motivation per change in write scores do not. If relative changes also differ, we include interaction terms.

Interaction effects

Interaction effects exist when the effect of an independent variable on a dependent variable changes, depending on the value of another independent variable e.g the relationship between age, smoking and risk of lung-cancer. Smoking increases risk of lung-cancer (affects intercept). Age is associated with increased risk (affects slope). However, there is an elevated risk that comes from the combination of being older and smoking - i.e. an interaction (affects slope). For the motivation example we can add interaction terms for write and program_ as shown in Figure 5:

```
> mod3 <- lm(motivation ~ write*program_)
> summary(mod3)

Call:
lm(formula = motivation ~ write * program_)

Residuals:
    Min       1Q   Median       3Q      Max
-44.30  -9.28  -0.51   9.67  42.83

Coefficients:
            Estimate Std. Error t value
(Intercept)  29.58902   4.60073   6.431
write         0.42236   0.08678   4.867
program_general -0.53380   7.96402  -0.067
program_vocational 12.82973   7.79651   1.646
write:program_general -0.12615   0.15269  -0.826
write:program_vocational -0.15129   0.14409  -1.050
---
```

Figure 5. Interaction effect example

Including the interaction effects changes the interpretation of our coefficients:

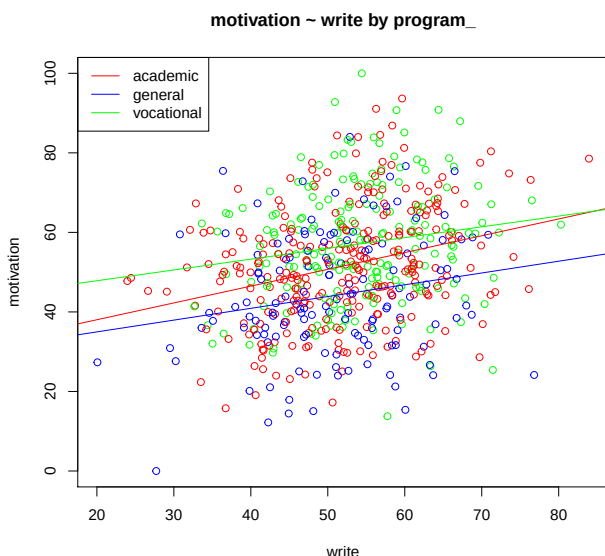
- $\hat{\beta}_0 = 29.59 \implies$ The expected motivation score for a student who obtained 0 for their write score and who is in the academic program is 29.59.
- $\hat{\beta}_{\text{write}} = 0.42 \implies$ The average increase in the motivation score per unit increase in the write score is 0.42 for students in the academic program.
- $\hat{\beta}_{\text{program_general}} = -0.53 \implies$ The average decrease in the motivation score comparing general to academic program students is 0.53, holding the write score constant.

- $\hat{\beta}_{\text{program_vocational}} = 12.83 \implies$ The average increase in the motivation score comparing vocational to academic program students is 12.83, holding the write score constant.
- $\hat{\beta}_{\text{write:program_general}} = -0.13 \implies$ The average decrease in the coefficient for write when comparing general to academic program students is 0.13.
- $\hat{\beta}_{\text{write:program_vocational}} = -0.15 \implies$ The average decrease in the coefficient for write when comparing general to academic program students is 0.15.

Prediction of motivation score (y) using write score (x) and academic program has 3 possible estimation equations for academic (1), general (2) and vocational (3) programs, respectively:

$$\begin{aligned} (1) \hat{y} &= 29.59 + 0.42 * x \\ (2) \hat{y} &= 29.59 + (0.42 - 0.13) * x - 0.53 \\ (3) \hat{y} &= 29.59 + (0.42 - 0.15) * x + 12.83 \end{aligned}$$

```
coef <- coefficients(mod3)
plot(write, motivation,
     main="motivation ~ write by program_",
     col=c("red", "blue", "green")[program_])
abline(coef[1], coef[2], col="red")
abline(coef[1]+coef[3], coef[2]+coef[5], col="blue")
abline(coef[1]+coef[4], coef[2]+coef[6], col="green")
legend("topleft",
     legend=c("academic", "general", "vocational"),
     col=c("red", "blue", "green"), lty=1)
```



Hypothesis Testing

As with simple linear regression, the test statistics required for hypothesis testing are linked to the Analysis of Variance (ANOVA) table shown in Figure 6.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression	p	$SSR = \sum_i (\hat{y}_i - \bar{y})^2$	$MSR = \frac{SSR}{p}$	$\frac{MSR}{MSE}$
Error	$n - p - 1$	$SSE = \sum_i (\hat{y}_i - y_i)^2$	$MSE = \frac{SSE}{n - p - 1}$	
Total	$n - 1$	$SST = \sum_i (y_i - \bar{y})^2$	$MST = \frac{SST}{n - 1}$	

Figure 6. Multiple Regression ANOVA

Test for overall significance of linear regression

The test statistic: $Fstat = \frac{MSR}{MSE} \sim F_{p, n-p-1}$ can be used to test for the significance of the regression based on the hypotheses:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$$

$$H_1 : \text{At least one } \beta_j \neq 0$$

As with SLR, we derive the Multiple R^2 defining the proportion of the variation in the data that is explained by the model, equals to SSR/SST . The R^2 always increases with increasing number of parameters despite the corresponding loss in degrees of freedom and power and so to adjust for this we have the adjusted R_a^2

$$R_a^2 = 1 - \frac{n-1}{n-p-1}(1-R^2) = 1 - \frac{MSE}{MST}$$

Tests for individual regression coefficients

The test statistic: $tval = \frac{\hat{\beta}_j}{s_{\beta_j}} \sim t_{n-p-1}$ can be used to test for the significance of each specific β_j for $j \in [0, p]$ based on the hypotheses,

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0$$

Statistical significance can also be ascertained using the $\beta_j 100(1 - \alpha)\%$ confidence intervals i.e.:

$$\beta_j \in \hat{\beta}_j \pm t_{\alpha/2, n-p-1} s_{\beta_j}$$

If interval excludes 0 $\implies$ **statistically significant!**


```

> mrmod2 <- lm(motivation ~ write + read + science + program_ + program_*read)
> summary(mrmod2)

Call:
lm(formula = motivation ~ write + read + science + program_ +
    program_* read)

Residuals:
    Min       1Q   Median       3Q      Max
-43.530  -9.292  -0.423   9.512  43.072

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  27.79192    4.80523   5.784 1.18e-08 ***
write         0.32783    0.07982   4.107 4.57e-05 ***
read         0.30441    0.10772   2.826 0.00487 **
science      -0.17815    0.08573  -2.078 0.03813 *
program_general  3.67859    7.82911   0.470 0.63863
program_vocational 17.58248    7.24163   2.428 0.01548 *
read:program_general -0.20399    0.15265  -1.336 0.18194
read:program_vocational -0.24126    0.13479  -1.790 0.07398 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.26 on 592 degrees of freedom
Multiple R-squared:  0.1554,    Adjusted R-squared:  0.1454
F-statistic: 15.56 on 7 and 592 DF,  p-value: < 2.2e-16

```

Figure 7. Final Multiple Regression in RStudio

Motivation scores: examples

We can predict the motivation score using the academic scores, the program and an interaction of program and read. Using RStudio we derive Figure 7. Two example questions we can use this output to answer,

- 1) Is there a significant interaction between read and program for a general program student?
- 2) Predict the motivation score of a vocational program student who obtained 60, 80 and 90 scores for write, read and science, respectively.

For (1), using a 5% significance level, the p-value of 0.18194 is sufficient evidence to **fail to reject** the null hypothesis

$$H_0 : \beta_{\text{read:program_general}} = 0$$

i.e. conclude the interaction is not statistically significant.

For (2) substitute relevant values into the estimation equation:

$$\begin{aligned}
 \hat{y} &= 27.79192 + 0.32783 * 60 \\
 &+ (0.30441 - 0.24126) * 80 - 0.17815 * 90 \\
 &+ 17.58248 \approx 54.06
 \end{aligned}$$

Comment: Multicollinearity

Multicollinearity occurs when independent variables in a regression model are correlated. This correlation is a problem because independent variables should be independent. If the degree of correlation between variables is high enough, it can cause problems when you fit the model and interpret the results. Multicollinearity causes the following two basic types of problems:

- The coefficient estimates can swing wildly based on which other independent variables are in the model. The coefficients become very sensitive to small changes in the model.
- Multicollinearity reduces the precision of the estimate coefficients, which weakens the statistical power of your regression model. You might not be able to trust the p-values to identify independent variables that are statistically significant.

In this course we do not correct for multicollinearity but we take note of the circumstances under which it occurs and how it may affect our estimation and inference.

4. Model building

So far we have assumed the explanatory variables included in the model are chosen in advance. However, in many situations the variables are not predetermined and selecting them is part of the analysis.

To describe the model selection approaches we will use the `mr` motivation dataset introduced earlier but edited slightly for illustrative purposes. The potential variables we will look at are shown in the Figure 8 linear model. A streamlined dataset `mrmb` is derived by manually creating dummy variables and restricting data for analysis to specific variables:

```
> sm <- lm(motivation ~ read + write + program_, data = mr)
> summary(sm)

Call:
lm(formula = motivation ~ read + write + program_, data = mr)

Residuals:
    Min       1Q   Median       3Q      Max
-44.305  -9.658  -0.814   9.710  43.386

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    31.59789    3.55868   8.879 < 2e-16 ***
read           0.09179    0.07490   1.225 0.220887
write          0.29132    0.07767   3.751 0.000194 ***
program_general -6.83279    1.50820  -4.530 7.12e-06 ***
program_vocational 4.86971    1.35891   3.584 0.000367 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.32 on 595 degrees of freedom
Multiple R-squared:  0.1444,    Adjusted R-squared:  0.1387
F-statistic: 25.11 on 4 and 595 DF,  p-value: < 2.2e-16

> mrmb <- mr
> mrmb$pr_gn <- ifelse(mrmb$program_=="general",1,0)
> mrmb$pr_vc <- ifelse(mrmb$program_=="vocational",1,0)
> mrmb <- mrmb[, c("motivation","read","write","pr_gn","pr_vc")]
> |
```

Figure 8. Full model for selection

To compare models we calculate the Akaike information criterion (AIC) for each

$$AIC = n * \log(SSE/n) + 2p + 2$$

which can be used to assess if a model is optimal. The AIC estimates the relative quality of statistical models for a given set of data Y such that the model with the **lowest** AIC is deemed to be the best. Utilizing the AIC, we focus on three variable selection approaches: the **all possible regressions approach**, and the **forward selection** and **backward elimination** procedures.

All possible regressions

This approach considers all possible subsets of the pool of explanatory variables and finds the model that best fits the data according to the lowest AIC. If there are p variables under consideration, there are 2^p possible models, e.g. for 4 variables: `read`, `write`, `pr_gn` and `pr_vc`, we have $2^4 = 16$ possible models (including constant model):

Table 2. All subsets regression

(Intercept)	read	write	pr_gn	pr_vc	adjR2	AIC
33.311		0.351	-7.034	4.801	0.138	3,198.368
31.598	0.092	0.291	-6.833	4.870	0.139	3,198.856
34.325		0.369	-8.982		0.121	3,208.815
32.831	0.081	0.317	-8.829		0.122	3,209.668
37.551	0.267		-6.666	5.342	0.120	3,210.875
27.931	0.129	0.281		7.144	0.110	3,217.204
30.208		0.365		7.141	0.108	3,218.109
39.497	0.272		-8.858		0.099	3,224.035
33.763	0.297			7.546	0.093	3,227.834
51.575			-7.465	5.475	0.091	3,229.167
53.839			-9.728		0.069	3,242.567
28.205	0.128	0.320			0.066	3,245.615
30.475		0.403			0.063	3,246.357
49.057				7.994	0.057	3,250.381
34.908	0.322				0.043	3,259.231
51.601					0	3,284.453

Our results suggest the best model has an AIC of 3198.37. It differs from the full/saturated model in that the variable `read` is not included in it. Looking at Figure 9 output, the full model would have been chosen based on R^2 or $\text{adj}R^2$ for which it is higher for both. This contradicts the choice of best model when decision is based on the AIC.

```
> summary(lm(motivation ~ write + program_))

Call:
lm(formula = motivation ~ write + program_)

Residuals:
    Min       1Q   Median       3Q      Max
-44.602  -9.285  -0.621   9.717  42.796

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    33.31112    3.27395  10.175 < 2e-16 ***
write           0.35086    0.06063   5.787 1.16e-08 ***
program_general -7.03377    1.49989  -4.690 3.40e-06 ***
program_vocational 4.80116    1.35833   3.535 0.00044 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.32 on 596 degrees of freedom
Multiple R-squared:  0.1423,    Adjusted R-squared:  0.138
F-statistic: 32.95 on 3 and 596 DF,  p-value: < 2.2e-16
```

Figure 9. Selected model from All subsets

Forward Selection

Forward selection starts with a small model, e.g. just an intercept, considers all one-variable expansions of the model, and retains the variable which creates a model that is best according to the AIC. This process is repeated, adding one variable at a time, until the AIC stops falling any further.

```
constant <- lm(motivation ~ 1, data=mrmb)
full <- lm(motivation ~ ., data=mrmb)

step(constant, direction = "forward",
scope = list(lower=constant, upper=full))

## Start: AIC=3284.45
## motivation ~ 1
##      Df Sum of Sq  RSS   AIC
## + pr_gn  1   10056.7 132527 3242.6
## + write  1    9216.9 133367 3246.4
## + pr_vc  1    8319.6 134264 3250.4
## + read   1    6324.5 136259 3259.2
## <none>                142584 3284.5
##
## Step: AIC=3242.57
## motivation ~ pr_gn
##      Df Sum of Sq  RSS   AIC
## + write  1    7666.2 124861 3208.8
## + read   1    4458.5 128069 3224.0
## + pr_vc  1    3358.4 129169 3229.2
## <none>                132527 3242.6
##
## Step: AIC=3208.82
## motivation ~ pr_gn + write
##      Df Sum of Sq  RSS   AIC
## + pr_vc  1    2563.60 122297 3198.4
## <none>                124861 3208.8
## + read   1     238.63 124622 3209.7
##
## Step: AIC=3198.37
## motivation ~ pr_gn + write + pr_vc
##      Df Sum of Sq  RSS   AIC
## <none>                122297 3198.4
## + read   1     307.89 121989 3198.9
##
## Call:
## lm(formula = motivation ~ pr_gn + write + pr_vc,
##     data = mrmb)
##
## Coefficients:
## (Intercept)      pr_gn      write      pr_vc
##      33.3111     -7.0338      0.3509      4.8012
```

In the first step, the forward selection starts out with the constant model $\text{motivation} = \beta_0$ and calculates its AIC to be 3284.45. It then calculates AIC for all one variable models i.e.

$$\text{motivation} = \beta_0 + \text{pr_gn} \text{ with AIC } 3242.6$$

$$\text{motivation} = \beta_0 + \text{write} \text{ with AIC } 3246.4$$

$$\text{motivation} = \beta_0 + \text{pr_vc} \text{ with AIC } 3250.4$$

$$\text{motivation} = \beta_0 + \text{read} \text{ with AIC } 3259.2$$

The $\text{motivation} = \beta_0 + \text{pr_gn}$ model is the best with lowest AIC of 3242.57 and it becomes the new starting point onto which second variables are added i.e.

$$\text{motivation} = \beta_0 + \text{pr_gn} + \text{write} \text{ with AIC } 3208.8$$

$$\text{motivation} = \beta_0 + \text{pr_gn} + \text{read} \text{ with AIC } 3224.0$$

$$\text{motivation} = \beta_0 + \text{pr_vc} + \text{write} \text{ with AIC } 3229.2$$

This process is repeated until there is no addition of a variable that would lead to a lower AIC. The chosen model is the same as that from the All subsets regression, with all variables except read..

Backward Elimination

Contrary to forward selection, in backward elimination, we start with the largest model we're willing to contemplate, and keep eliminating variables until we no longer reduce the AIC.

```
step(full, direction = "backward",
scope = list(lower=constant, upper=full))

## Start: AIC=3198.86
## motivation ~ read + write + pr_gn + pr_vc
##      Df Sum of Sq  RSS   AIC
## - read   1     307.9 122297 3198.4
## <none>                121989 3198.9
## - pr_vc  1    2632.9 124622 3209.7
## - write  1    2884.0 124873 3210.9
## - pr_gn  1    4208.1 126197 3217.2
##
## Step: AIC=3198.37
## motivation ~ write + pr_gn + pr_vc
##      Df Sum of Sq  RSS   AIC
## <none>                122297 3198.4
## - pr_vc  1    2563.6 124861 3208.8
## - pr_gn  1    4512.6 126810 3218.1
## - write  1    6871.4 129169 3229.2
##
## Call:
## lm(formula = motivation ~ write + pr_gn + pr_vc,
##     data = mrmb)
##
## Coefficients:
## (Intercept)      write      pr_gn      pr_vc
##      33.3111      0.3509     -7.0338      4.8012
```

For this example, at step 1 the AIC is calculated for the full model and found to be 3198.86. Only removing read leads to a lower AIC of 3198.37, otherwise there is no other removal that improves our modelling. And so the model without read is the selected model.

5. Logistic regression

Introduction

Logistic regression is a statistical method for analyzing a dataset in which the dependent variable is binary/dichotomous i.e. it only has two possible outcomes. As with linear regression, we still have one or more independent variables to determine the extent to which they are associated with the outcome. Our objectives in the logistic regression section are to learn:

1. The appropriate mathematical model for estimating the regression relationship.
2. How to estimate the model parameters using RStudio .
3. How to interpret the meanings of the estimated parameters and test them for statistical significance.
4. How to make predictions given new values of the explanatory variables.
5. How to assess model performance by determining the sensitivity and specificity as well as positive and negative predictive values in a confusion matrix.

Modelling binary data

Let's define a binary response variable Y which is coded to 1 if some characteristic of interest is present (e.g. TRUE, success, pregnant, alive, etc) or 0 if the characteristic is absent (e.g. FALSE, failure, non-pregnant, dead, etc). We are interested in the probability that this characteristic is present i.e.

$$Pr(\text{Characteristic present}) = Pr(Y = 1) = p$$

and how this probability is associated with one or more explanatory variables.

A linear model of the form $p = \beta_0 + \beta_1 x$ will be unsuitable because if the β_j are free to take on any value between $(-\infty, +\infty)$ then a linear combination of the β_j can take on any value over that range. We off-course want p to lie within the interval $(0,1)$. To ensure this, we use the LOGIT transformation which generates a new variable

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

which becomes our dependent variable in a linear model consisting of the explanatory variables,

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

The ratio of p against $(1-p)$ are the **ODDS** i.e.

$$\text{odds} = \frac{p}{1-p} = \frac{\Pr(\text{Characteristic present})}{\Pr(\text{Characteristic absent})}$$

and we note that our logistic regression model predicts the **LOG-ODDS** of the characteristic being present.

A logistic regression model

As an example we look at the low birthweight dataset.² The goal of the study was to identify risk factors for low birthweight using data on 189 women, 59 of which had low birthweight babies. We model the outcome `lowbwt` (1 for low and 0 for normal) using the mothers weight `lwt`, the mother's race (categories black, other and white) and the variable `smoke` (categories no and yes) for smoking status. The data are imported into RStudio :

```
lowbwt <- read.table("lowbwt.txt", header = T)
head(lowbwt)
```

##	lowbwt	age	lwt	race	smoke	ui	ftv
## 1	1	28	120	Other	Yes	Yes	None
## 2	1	29	130	White	No	Yes	Two, etc.
## 3	1	34	187	Black	Yes	No	None
## 4	1	25	105	Other	No	No	None
## 5	1	25	85	Other	No	Yes	None
## 6	1	27	150	Other	No	No	None

The logistic regression model predicts the log-odds of a child having a low birthweight as a function of the explanatory variables i.e.

$$\log\left(\frac{\Pr(\text{lowbwt} = 1)}{\Pr(\text{lowbwt} = 0)}\right) \sim f(\text{race}, \text{smoke}, \text{lwt})$$

Although the dependent variable in the syntax is `lowbwt`, the `glm` command with the binomial specification tell RStudio that we want to model the log-odds.

²Hosmer and Lemeshow (2000) Applied Logistic Regression: Second Edition

```

> logmod <- glm(lowbwt ~ lwt + race + smoke, data = lbwt, family = "binomial")
> summary(logmod)

call:
glm(formula = lowbwt ~ lwt + race + smoke, family = "binomial",
    data = lbwt)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.5278  -0.9053  -0.5863   1.2878   2.0364

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.18087    1.00983   1.169  0.24225
lwt          -0.01326    0.00631  -2.101  0.03562 *
raceOther    -0.31958    0.52560  -0.608  0.54317
raceWhite   -1.29009    0.51087  -2.525  0.01156 *
smokeYes      1.06001    0.37832   2.802  0.00508 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 10. Logistic regression in RStudio

Interpreting coefficients

Coefficients are interpreted in terms of the log-odds and can be exponentiated to interpret them in terms of the odds. For this example:

- $\beta_0 = 1.18 \implies$ the **log-odds** of a child being of low birthweight if born to a non-smoking black mother of weight 0 is 1.18. The **odds** of a child being of low birthweight if born to a non-smoking black mother of weight 0 is $e^{1.18} = 3.25$.
- $\beta_{lwt} = -0.01326 \implies$ holding race and smoking constant, there is an average decrease of 0.01326 in the log-odds of a child being of low birthweight per one pound increase in the mother's weight. This implies that the odds are scaled by a factor of $e^{-0.013} = 0.987$, or equivalently, the odds decrease by $100 - 98.7 = 1.3\%$, per pound increase in mother's weight, holding race and smoking constant.
- $\beta_{raceOther} = -0.32 \implies$ there is an average decrease of 0.32 in the log-odds of a child being of low birthweight if the mother is of race "Other" vs if she is black, holding smoking status and mother's weight constant. We can also say the odds are scaled by a factor of $e^{-0.32} = 0.73$, or equivalently, the odds decrease by a $100 - 73 = 27\%$ if the mother is of race "Other" vs if she is black. This is an **odds-ratio of 0.73**, comparing race other to race black.
- $\beta_{raceWhite} = -1.29 \implies$ holding smoking status and mother's weight constant, there is an average decrease of 1.29 in the log-odds of a child being of low birthweight if the mother is white vs if she is black. In other words, the odds are scaled by a factor of $e^{-1.29} = 0.28$, or decrease by $100 - 28 = 72\%$ if the mother is white vs if she is black. This is an odds-ratio of 0.28.
- $\beta_{smokeYes} = 1.06 \implies$ holding race and mother's weight constant, there is an average increase of 1.06 in the log-odds of a child being of low birthweight if the mother smokes vs if she does not. This implies an increase in the odds by a factor of $e^{1.06} = 2.89$, or equivalently an increase of $289 - 100 = 189\%$. As defined for other coefficients, 2.89 is the **odds-ratio**, comparing the odds of being low-birthweight for children born to mother's who smoke to the odds of being low-birthweight for those born to mother's who do not.

Inference

The coefficients and their standard errors are used to estimate test statistics that can be used to perform inference. We can test the significance of β_j ,

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0$$

using the test statistic:

$$z\text{-val} = \frac{\hat{\beta}_j}{s_{\beta_j}} \sim N(0, 1)$$

which follows a standard normal distribution. If the p-value is sufficiently small, we reject H_0 and conclude a statistically significant relationship. For example, if we are wanting to test if the coefficient β_{smokeYes} is statistically significant, we can use

$$z\text{-val} = \frac{1.06001}{0.37832} = 2.802$$

Our p-value, which is the probability of observing a value greater than 2.802 on the standard normal distribution, is 0.00508. This is sufficiently small, we reject H_0 and conclude that mother's smoking increases the risk of a child having a low birthweight. The 95% confidence interval for β_j is:

$$\hat{\beta}_j \pm 1.96 \times s.e.(\hat{\beta}_j)$$

If the interval excludes 0 $\implies$ **statistically significant**. For example, the 95% CI for β_{smokeYes} is:

$$\beta_{\text{smokeYes}} \in 1.06 \pm 1.96 * 0.378 = [0.319, 1.802]$$

This means we are 95% sure our true value of this coefficient lies between 0.32 and 1.8, an interval excluding 0 which implies a statistically significant coefficient. The 95% confidence interval for the odds can be estimated by exponentiating the lower and upper bounds of the log-odds 95% CI. For smoking odds-ratio, the 95% CI is

$$[e^{0.319}, e^{1.802}] = [1.38, 6.06]$$

This means that we are 95% sure that the true value of the odds-ratio lies between 1.38 and 6.06. This interval excludes 1 and implies a statistically significant odds-ratio.

Prediction

The sample estimates can be used to predict the logg-odds, the odds as well as the probability given specific values for the explanatory variables. For example, we may want to predict the probability that a child has a low birthweight if born to a 150 pound white mother who smokes:

$$\begin{aligned} \text{Log.Odds} &= 1.18087 - 0.01326 * 150 - 1.29009 \\ &\quad + 1.06001 = -1.03821 \\ \text{Odds} &= e^{-1.03821} = 0.3540879 \\ p &= \frac{\text{Odds}}{1 + \text{Odds}} = \frac{0.3540879}{1.3540879} = 0.26 \end{aligned}$$

In a similar way we can estimate the probability that a child has a low birthweight if born to a 150 pound black mother who does not smoke:

$$\begin{aligned} \text{Log.Odds} &= 1.18087 - 0.01326 * 150 = -0.80813 \\ \text{Odds} &= e^{-0.80813} = 0.4456907 \\ p &= \frac{\text{Odds}}{1 + \text{Odds}} = \frac{0.4456907}{1.4456907} = 0.31 \end{aligned}$$

Some of the code in Figure 11 is beyond the scope of the course but it illustrates how we can use a model consisting of just the mother's weight and her race to predict the probabilities of having a low birthweight.

```
> sm <- data.frame(lwt = rep(seq(0,300,1),3),
+ race = rep(c("Black", "Other", "white"), each = 301))
>
> lmod2 <- glm(lowbwt ~ race + lwt, data = lwt,
+ family = "binomial")
> summary(lmod2)

Call:
glm(formula = lowbwt ~ race + lwt, family = "binomial", data = lwt)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.3491  -0.8919  -0.7196   1.2526   2.0993

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.886820   0.998853   1.889  0.0589 .
raceOther    -0.600463   0.508931  -1.180  0.2381
raceWhite    -1.081066   0.488052  -2.215  0.0268 *
lwt          -0.015223   0.006439  -2.364  0.0181 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 234.67  on 188  degrees of freedom
Residual deviance: 223.26  on 185  degrees of freedom
AIC: 231.26

Number of Fisher Scoring iterations: 4

> sm$p <- predict(lmod2, newdata = sm, type = "response")
```

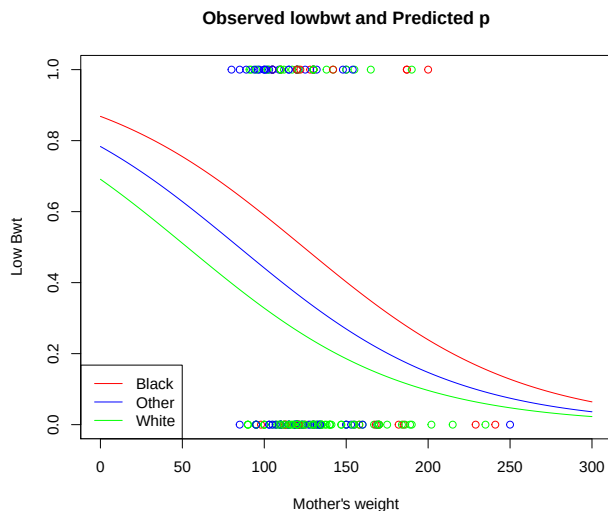
Figure 11. Race and weight model

Based on the results detailed in Figure 11, we can plot the probabilities of a child having a low birthweight and how these relate to the weight of the mother as well as her race.


```
plot(lbwt$lowbwt, lbwt$lowbwt, xlim = c(0,300),
     col=c("red", "blue", "green")[lbwt$race],
     xlab = "Mother's weight", ylab = "Low Bwt",
     main = "Observed lowbwt and Predicted p")

with(sm[sm$race=="Black",], lines(lwt, p, col="red"))
with(sm[sm$race=="Other",], lines(lwt, p, col="blue"))
with(sm[sm$race=="White",], lines(lwt, p, col="green"))

legend("bottomleft",
      legend=c("Black", "Other", "White"),
      col=c("red", "blue", "green"), lty=1)
```



At $lwt = 0$ we have the following probabilities:

- $p = \frac{1}{1+e^{-1.886820}} = 0.868$ for Black
- $p = \frac{1}{1+e^{-(1.886820-0.600463)}} = 0.784$ for Other
- $p = \frac{1}{1+e^{-(1.886820+0.081066)}} = 0.691$ for White

The predicted probabilities at every other value for the mother's weight can be calculated the same way.

The Confusion matrix

Ultimately, the logistic regression can be used to classify each observation to a 0 or 1 based on the predicted probability. We would like to assess the performance of the logistic classification. A confusion matrix is a table that compares the prediction classifications to the actual known values.

For the scenario resulting in Figure 12, our outcome is the presence of a disease (coded 1) or its absence. Once we have modeled this outcome using a logistic regression model and have classified observations to having the disease or not having it based on the probabilities of having the disease, we can define the following confusion matrix terms:

- true positives (TP):** Cases predicted as having the disease who actually have the disease.
- true negatives (TN):** Cases predicted no disease who actually don't have the disease.
- false positives (FP):** Predicted yes, but they actually don't have the disease (Type I error).
- false negatives (FN):** Predicted no, but they actually have the disease (Type II error).

	Disease Present	No Disease	
Positive Test	a (true positive)	b (false positive, Type 1 error)	PPV = $a/(a+b)$
Negative Test	c (false negative, Type 2 error)	d (true negative)	NPV = $d/(d+c)$
Sensitivity = $a/(a+c)$		Specificity = $d/(d+b)$	
PPV = $\frac{\text{true positives} \times 100\%}{(\text{true positives} + \text{false positives})}$		NPV = $\frac{\text{true negatives} \times 100\%}{(\text{true negatives} + \text{false negatives})}$	

Figure 12. Confusion matrix

Based on these, we can calculate the sensitivity, specificity, negative and positive predictive values of a model e.g. given the following confusion matrix where the outcome of interest was a "Yes":

		Actual attribute	
		Yes	No
Predicted attribute	Yes	85	42
	No	37	98

```
(Sens = 85/(85+37))
## [1] 0.6967213

(Spec = 98/(98+42))
## [1] 0.7

(PPV = 85/(85+42))
## [1] 0.6692913

(NPV = 98/(98+37))
## [1] 0.7259259
```